WEEK 4 CYBERSECURITY INTERNSHIP REPORT

Network Scanning, Web Server Access & Packet Analysis

Practical Lab Report

This report documents a controlled cybersecurity laboratory exercise using Kali Linux as the analysis/scanning machine and Metasploitable 2 as the intentionally vulnerable target. The work was performed on an isolated VirtualBox Host-Only network.

Lab Details

- Attacker / Analysis VM: Kali Linux — 192.168.56.101
- Target VM: Metasploitable 2 — 192.168.56.102
- Network: 192.168.56.0/24 (VirtualBox Host-Only)
- Primary tools: Nmap, Wireshark, Firefox, Linux networking commands
- Main activities: connectivity verification, Apache access, service discovery, packet capture and protocol filtering

The objective was to understand how a security analyst identifies reachable services and observes network traffic in a safe, authorized training environment.

1. Introduction

Cybersecurity assessment begins with understanding the network, identifying systems that are reachable, discovering exposed services, and observing the traffic generated by normal communication. This internship task provided hands-on practice with these activities in an isolated virtual laboratory.

Objectives

- Configure and verify communication between Kali Linux and Metasploitable 2.
- Access the target's Apache web server from Kali Linux.
- Use Nmap to identify open TCP ports and service versions.
- Capture and filter ICMP, HTTP, DNS and ARP traffic in Wireshark.
- Record observations that demonstrate why unnecessary or outdated services should be secured.
- Produce evidence that can be used in a cybersecurity internship report.

Ethical Scope

All scanning and packet analysis in this report was performed against the user's own intentionally vulnerable Metasploitable 2 virtual machine on an isolated Host-Only network. The techniques were used for authorized training and learning purposes only.

2. Lab Environment & Network Configuration

The practical environment consisted of two VirtualBox virtual machines. Kali Linux acted as the security testing and packet-analysis system, while Metasploitable 2 acted as the target system.

Network Configuration

- Kali Linux: 192.168.56.101/24
- Metasploitable 2: 192.168.56.102/24
- Subnet: 192.168.56.0/24
- Interface used for packet capture: eth0

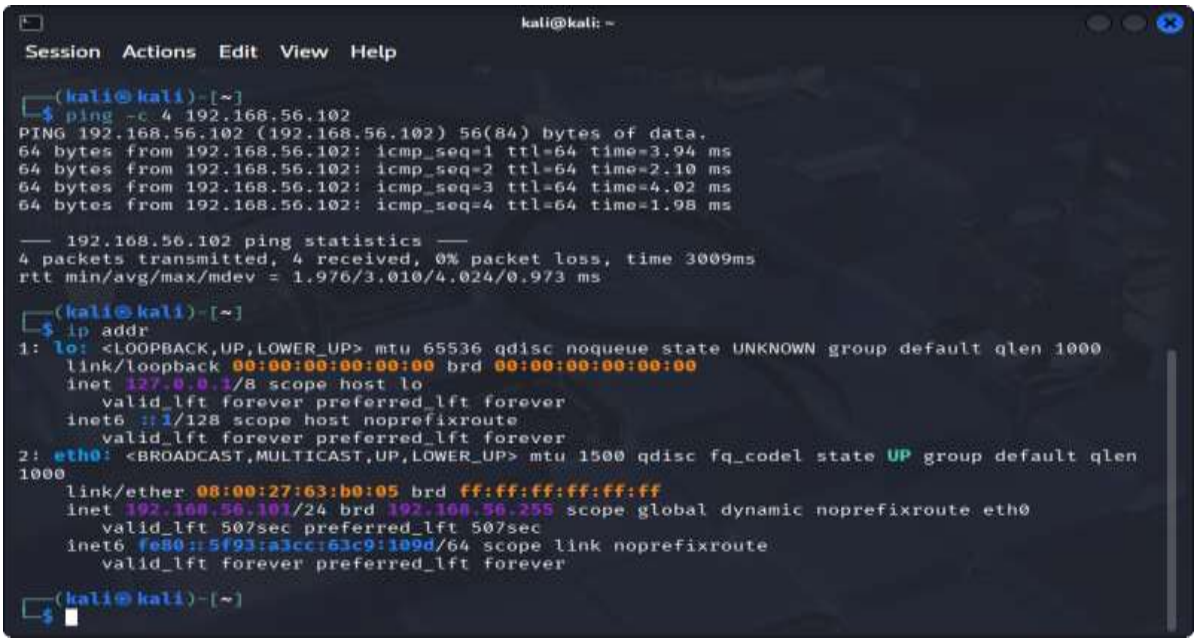


Figure 1. Kali Linux verifies connectivity to Metasploitable 2 and displays the eth0 address.

The ping test returned four replies with 0% packet loss, confirming that the two VMs could communicate over the isolated network.

3. Web Server Verification

Metasploitable 2 contains an Apache web server. The server was started on the target and accessed from Kali Linux using Firefox.

Procedure

- Start the Apache service on the Metasploitable 2 VM.
- Open Firefox in Kali Linux.
- Navigate to http://192.168.56.102.
- Verify that the Metasploitable web page is returned.
- Use the HTTP traffic generated by this access as Wireshark evidence.

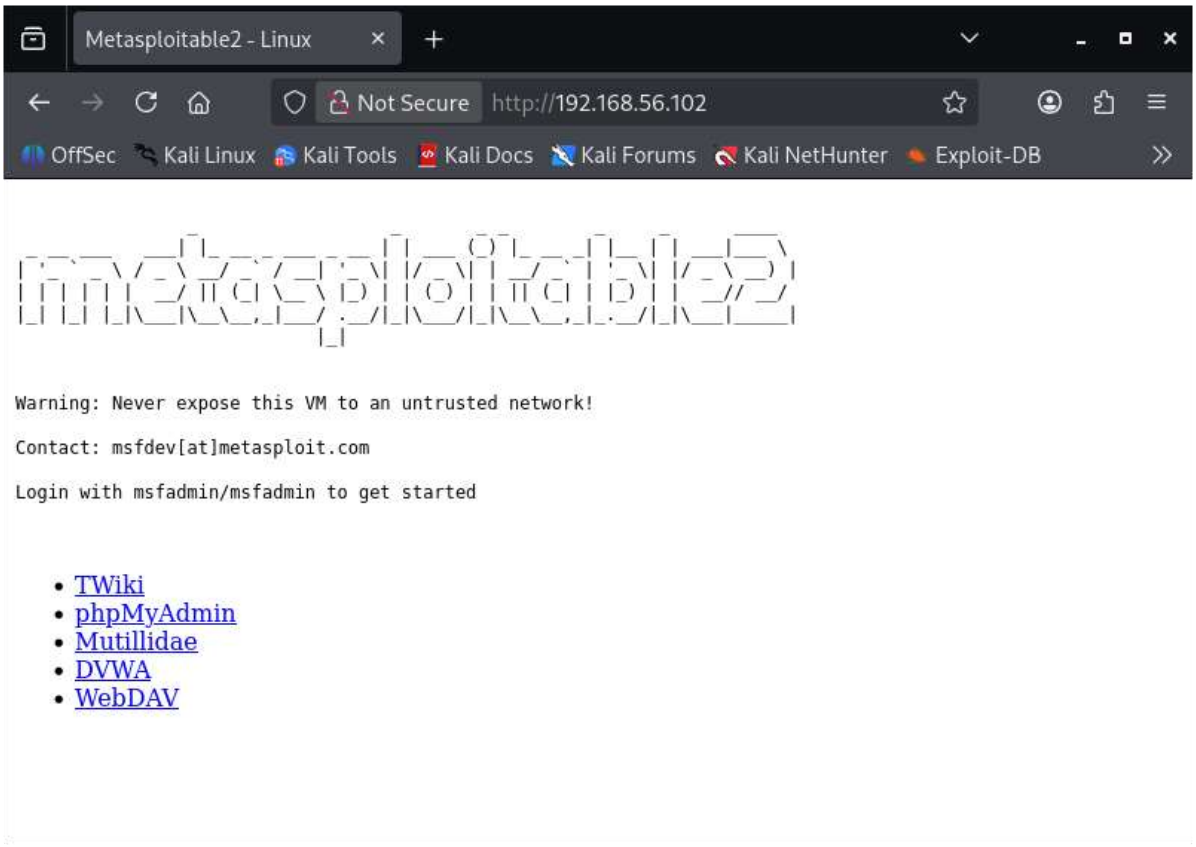


Figure 2. Metasploitable 2 Apache web page accessed from Kali Linux.

The successful page load confirmed that the target's HTTP service was reachable from the Kali VM.

4. Nmap Network Scanning

Nmap was used to enumerate TCP ports and identify services running on the target. A basic scan was followed by service/version detection.

Basic Scan

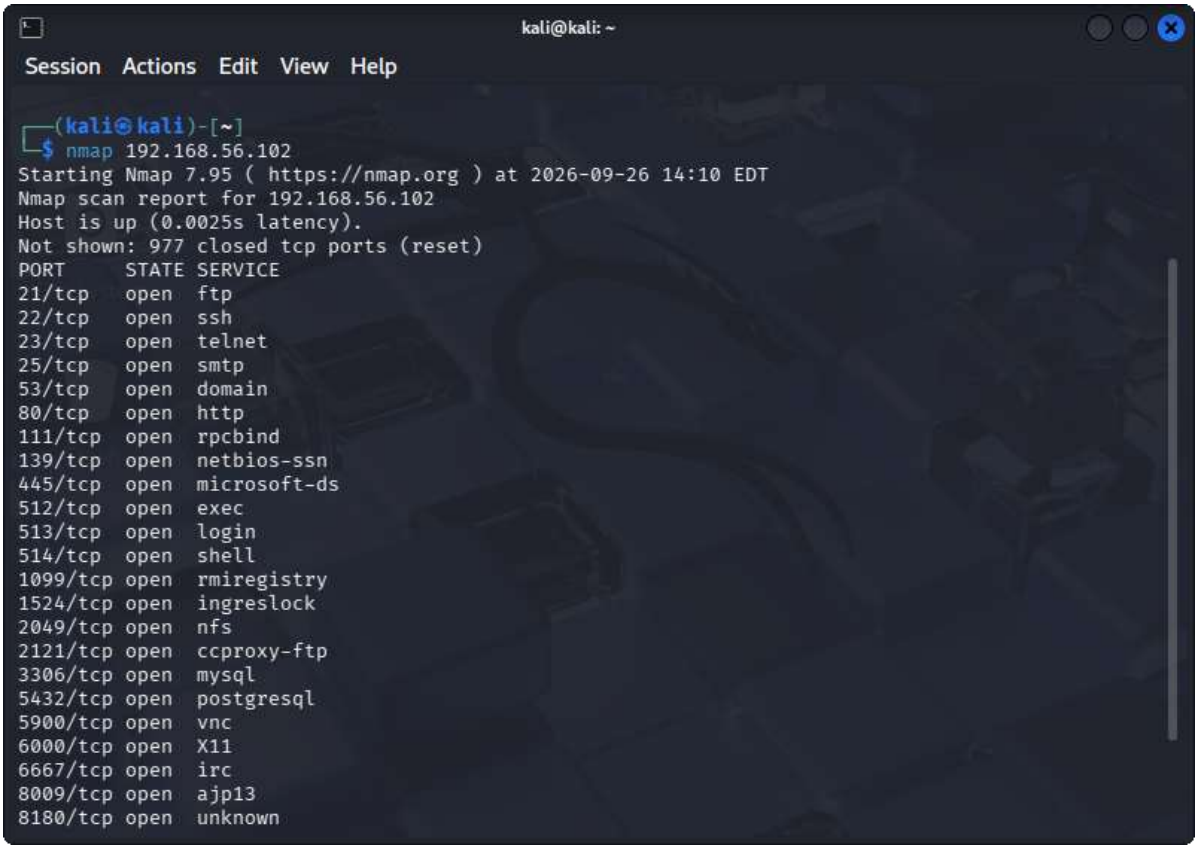


Figure 3. Basic Nmap scan of 192.168.56.102.

The basic scan identified numerous open TCP ports on the target. The scan reported 977 closed TCP ports and a set of open services.

Service and Version Detection

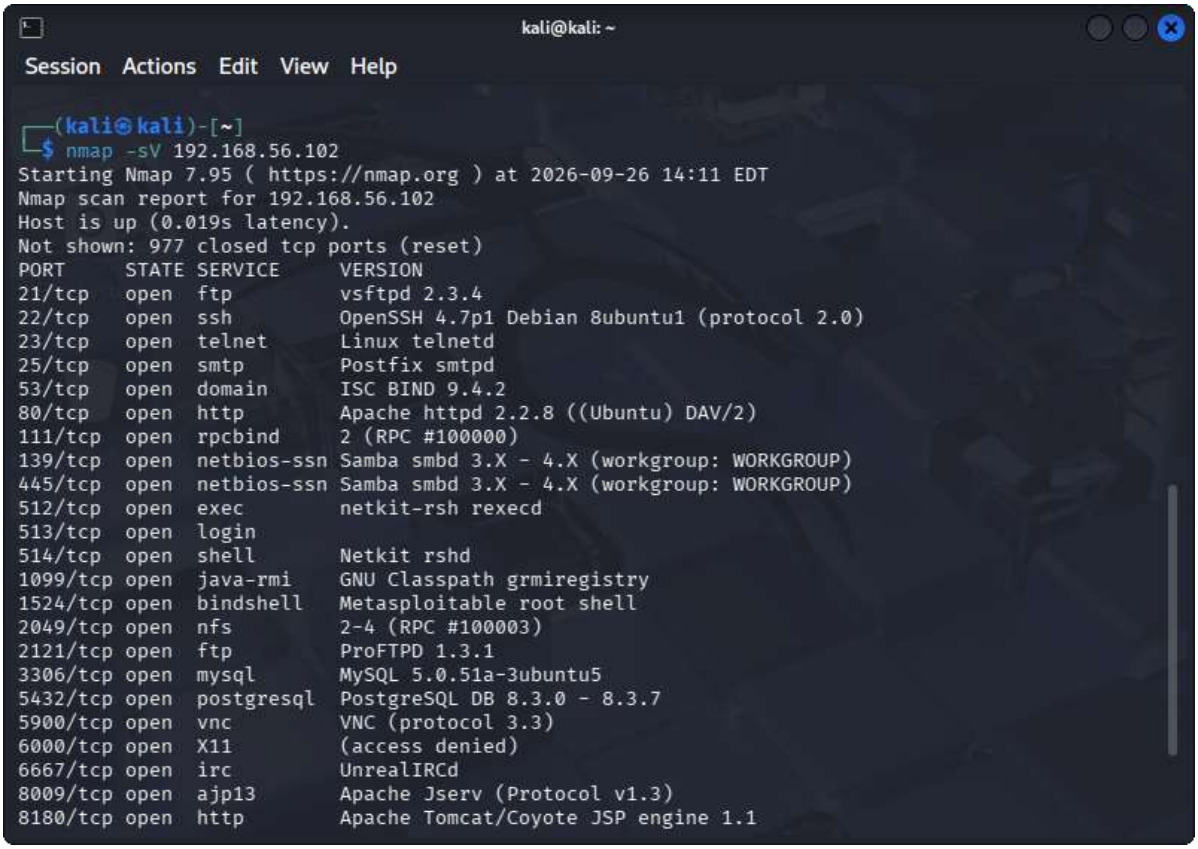


Figure 4. Nmap service/version detection (-sV).

The -sV scan provided service names and versions, making the results more useful for security assessment and later hardening decisions.

5. Nmap Findings & Service Inventory

The saved file week4_nmap_scan.txt contains the service/version results. The following inventory is based directly on the captured Nmap output.

Port	Service	Detected version
21	FTP	vsftpd 2.3.4
22	SSH	OpenSSH 4.7p1 Debian 8ubuntu1
23	Telnet	Linux telnetd
25	SMTP	Postfix smtpd
53	DNS	ISC BIND 9.4.2
80	HTTP	Apache httpd 2.2.8
139/445	NetBIOS/SMB	Samba 3.X–4.X
3306	MySQL	MySQL 5.0.51a
5432	PostgreSQL	PostgreSQL 8.3.0–8.3.7
5900	VNC	VNC protocol 3.3
8009	AJP13	Apache JServ
8180	HTTP	Apache Tomcat/Coyote JSP engine 1.1

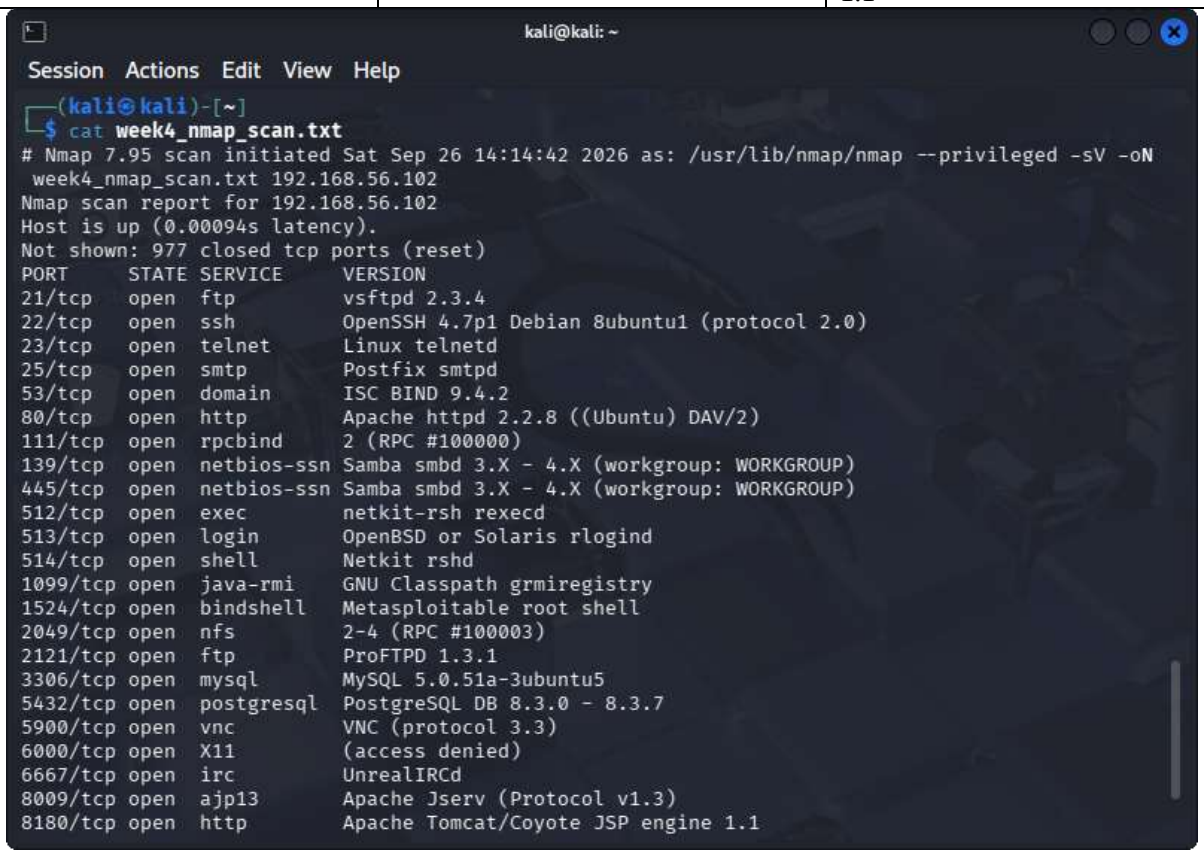


Figure 5. Nmap results saved to week4_nmap_scan.txt.

6. Wireshark – ICMP Traffic

Internet Control Message Protocol (ICMP) is commonly used for diagnostics such as ping. Wireshark was used with the display filter icmp while Kali generated traffic toward the target.

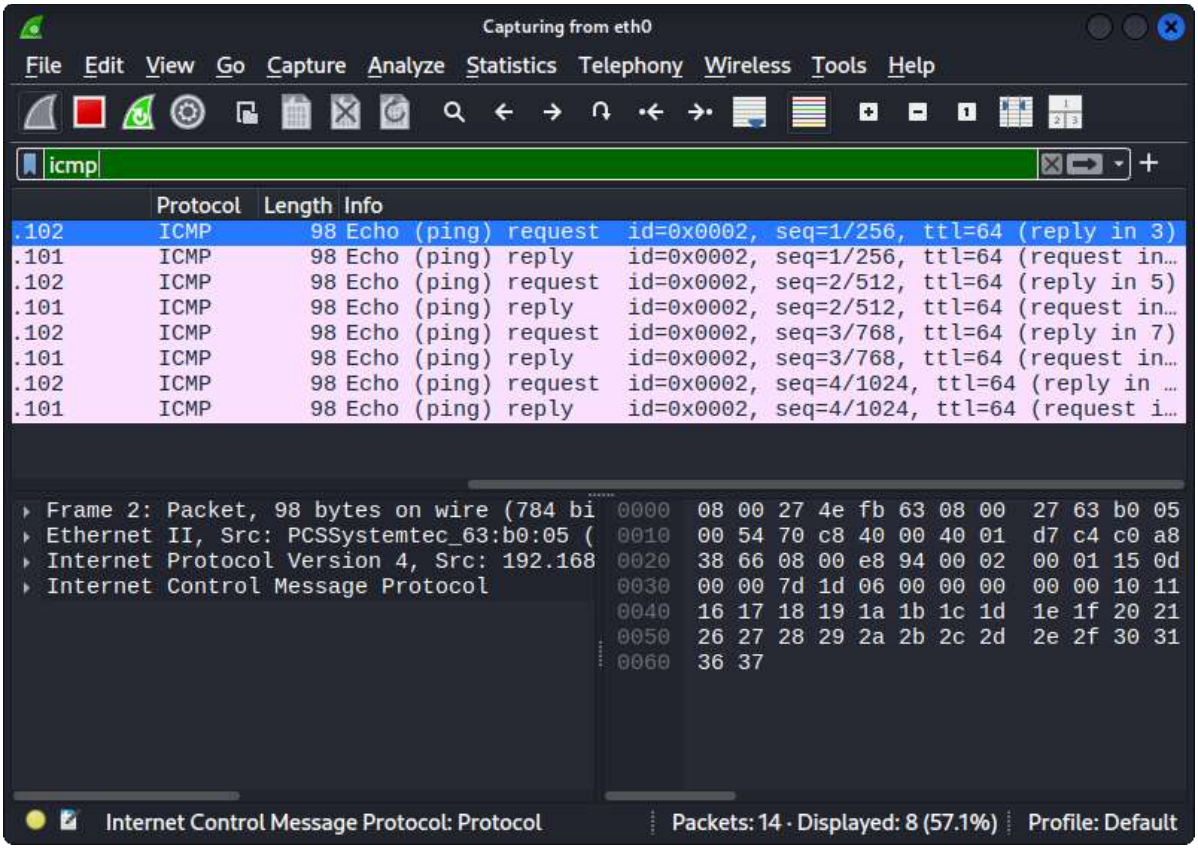


Figure 6. Wireshark ICMP filter showing Echo Request and Echo Reply packets.

Observation

- Echo Request packets were sent from Kali (192.168.56.101) to the target (192.168.56.102).
- Echo Reply packets were returned by the target.
- The packet exchange matches the successful ping test.
- The capture demonstrates how Wireshark can reveal source, destination and ICMP message type.

7. Wireshark – HTTP Traffic

HTTP traffic was captured while Kali accessed the target's Apache web page. The display filter http was used to isolate web traffic.

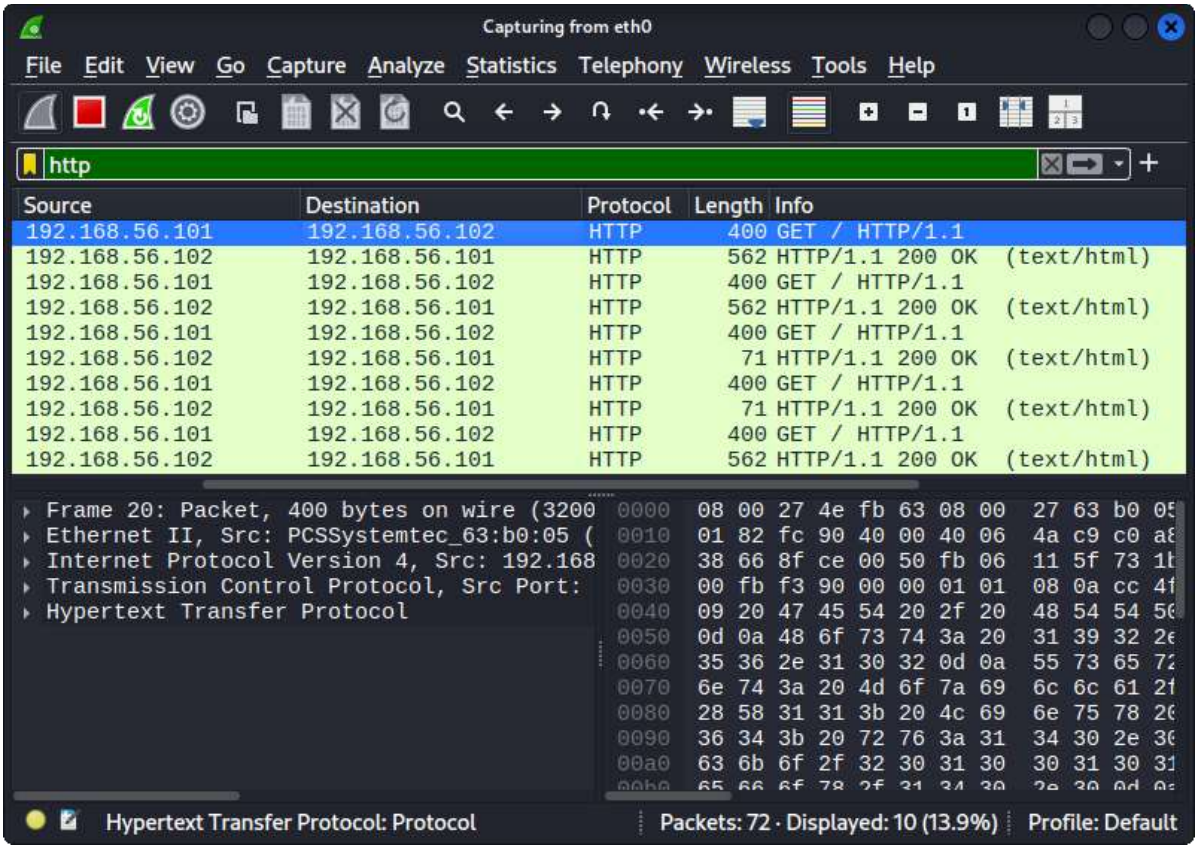


Figure 7. Wireshark HTTP filter showing GET requests and HTTP 200 OK responses.

Observation

- Kali generated HTTP GET requests toward 192.168.56.102.
- The target returned HTTP/1.1 200 OK responses.
- The capture confirms successful application-layer communication.
- Plain HTTP does not provide the same confidentiality protection as HTTPS, which is an important security consideration when sensitive information is transmitted.

8. Wireshark – ARP Traffic

Address Resolution Protocol (ARP) maps an IPv4 address to a MAC address on a local network. ARP traffic was captured on the Host-Only interface.

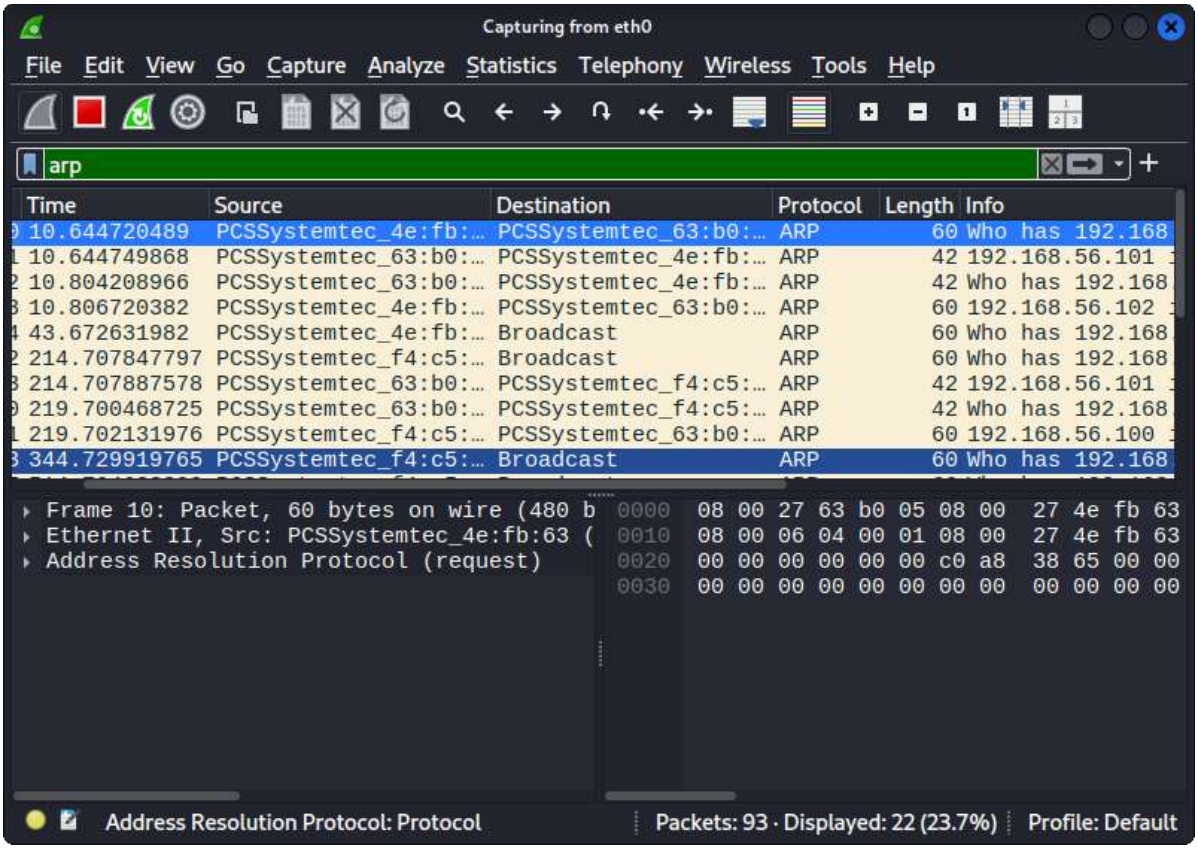


Figure 8. Wireshark ARP requests and replies on the 192.168.56.0/24 network.

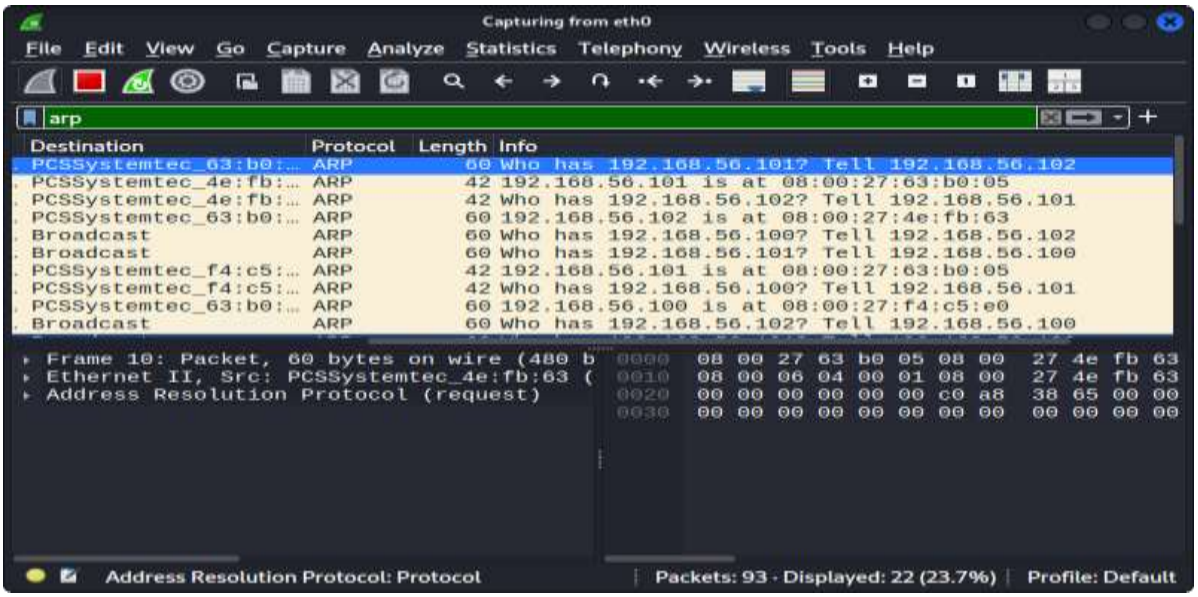


Figure 9. Detailed ARP request/reply traffic showing local address resolution.

Observation

The capture shows ARP request and reply messages used by hosts to determine the MAC address associated with a local IPv4 address. Because ARP is not authenticated, ARP spoofing is a known local-network security concern; network segmentation and monitoring can help reduce risk.

9. Wireshark – DNS Traffic

DNS traffic was captured using the display filter dns. Kali generated a DNS query toward the DNS service exposed by the target.

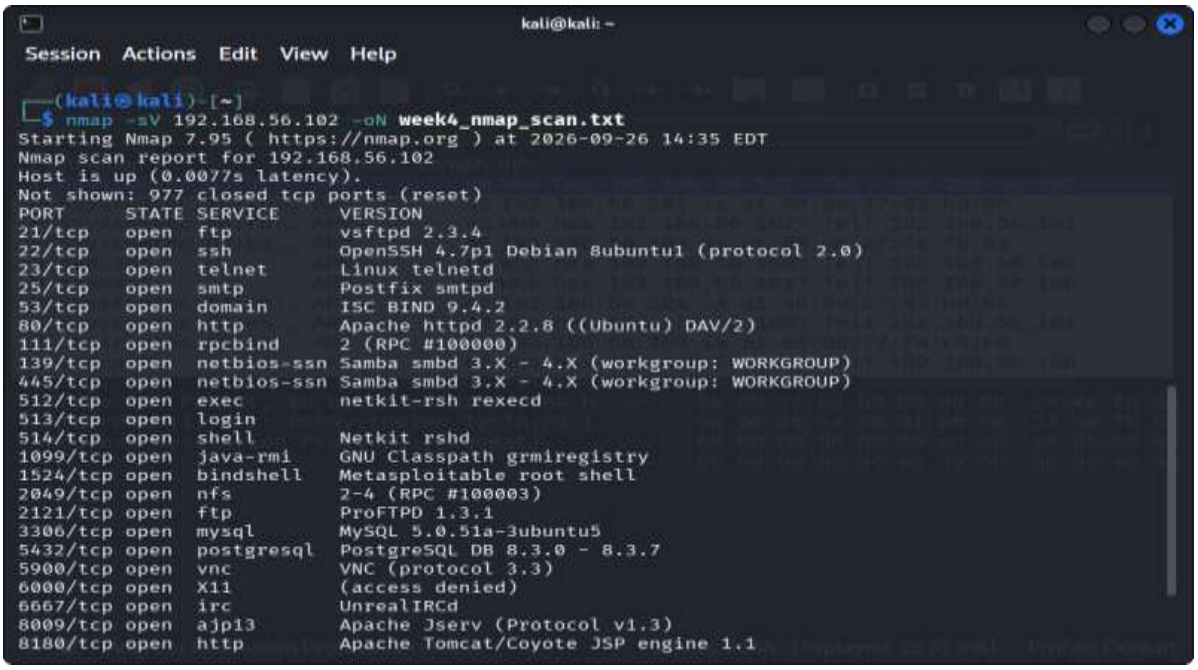


Figure 10. Wireshark DNS filter showing DNS query traffic.

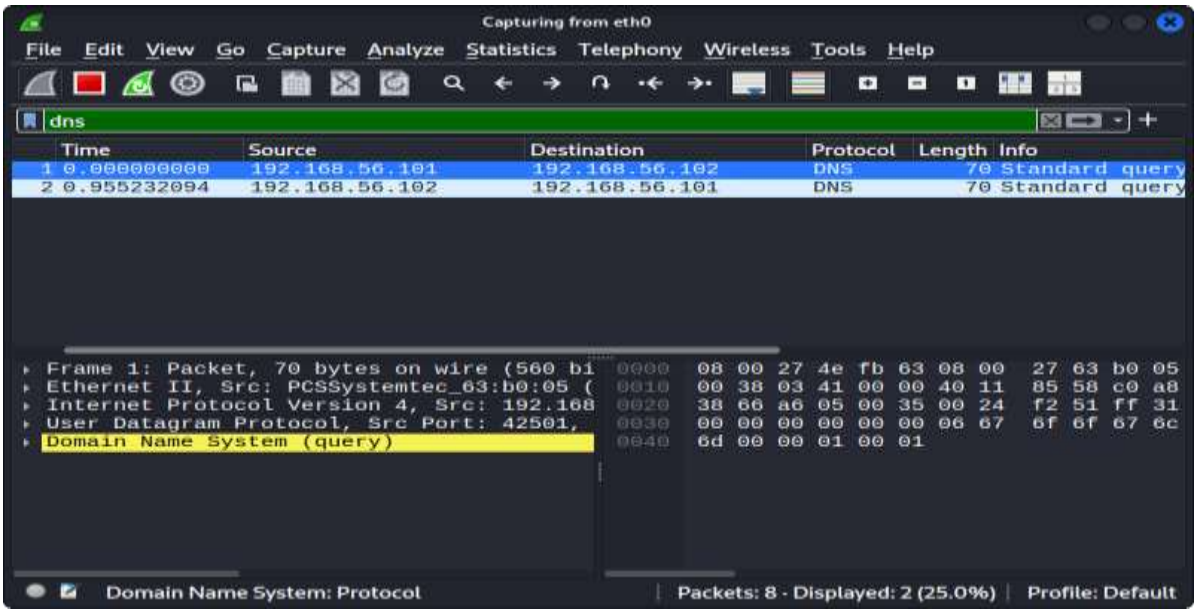


Figure 11. DNS query for google.com and the corresponding server-failure response.

Observation

- DNS packets were successfully captured between 192.168.56.101 and 192.168.56.102.
- The captured query requested an A record for google.com.
- The response shown in the capture indicates a server failure rather than a successful name resolution.
- This still demonstrates the ability to identify DNS requests and responses at the packet level.

10. Security Observations, Hardening & Conclusion

Security Observations

- The target exposes many network services, increasing its attack surface.
- Legacy services such as Telnet and older service versions should not be exposed unnecessarily.
- FTP and other clear-text protocols should be replaced with secure alternatives where possible.
- Only required services and ports should be enabled.
- Service versions should be kept current and unnecessary default services disabled.
- Firewall rules and network segmentation can restrict access to administrative and database services.
- Monitoring tools such as Wireshark can help analysts understand normal and abnormal traffic.

Basic Hardening Recommendations

- Disable unused services and close unnecessary ports.
- Use SSH instead of Telnet for remote administration.
- Use encrypted protocols such as HTTPS and secure file-transfer mechanisms where appropriate.
- Apply security updates and replace unsupported software versions.
- Restrict database, VNC, SMB and administrative services to trusted hosts or networks.
- Use host firewalls and network segmentation.
- Regularly scan authorized systems to verify the intended attack surface.

Conclusion

This Week 4 practical exercise provided hands-on experience with network connectivity testing, web-server verification, Nmap service discovery and Wireshark packet analysis. The lab demonstrated how a security analyst can move from basic connectivity verification to identifying exposed services and examining application and network protocols. The work was completed in an isolated VirtualBox environment using an intentionally vulnerable target, providing a safe foundation for further cybersecurity learning and interview preparation.

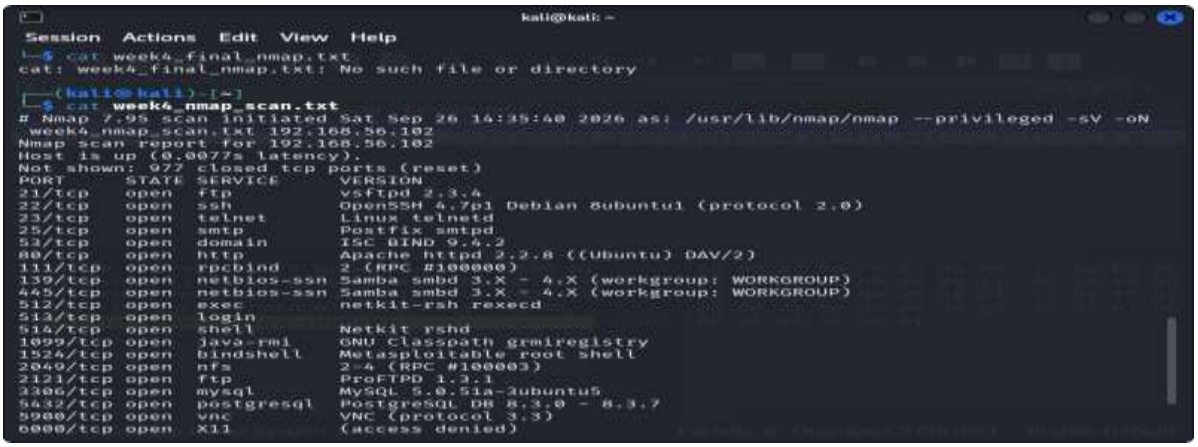


Figure 12. Final evidence: Nmap output viewed from the saved scan file.